

Hakob Atajyan

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EDUCATION

University of California, Los Angeles (UCLA)

Master of Science in Computer Science

Los Angeles, CA

Expected June 2028

- **Admission:** Exceptional Student Admissions Program (ESAP) Scholar; Early guaranteed admission for top undergraduates.

University of California, Los Angeles (UCLA)

Bachelor of Science in Computer Science

Los Angeles, CA

June 2026

- **GPA:** 3.86 | **Honors:** Dean's List (5x)
- **Relevant Coursework:** Data Structures and Algorithms, Software Construction, Computer Architecture, Computer Graphics, Data Mining, Operating Systems, Systems Architecture, Web Applications, Linear Algebra.

TECHNICAL PROJECTS

Voxel Destruction Simulator | [\[Live Demo\]](#) | *JavaScript, Three.js, WebGL*

Sept 2025 – Dec 2025

- Engineered a custom physics engine simulating rigid body dynamics, gravity, and radial impulses to model real-time structural collapse without external physics libraries.
- Achieved 60 FPS rendering of 10,000+ voxels by implementing **Geometry Instancing** and a dynamic state-swapping algorithm that converts static geometry to interactive debris upon impact.
- Optimized collision detection to O(1) by developing a spatial hashing system using coordinate-mapped lookups, significantly reducing computational overhead during high-density explosion events.
- Built a custom level parser to interpret raw text-based coordinate data and programmatically generate optimized 3D voxel grids with automated color mapping.

LoopZoo | [\[Live Deployment\]](#) | *Node.js, React, Express, MongoDB, Socket.IO*

April 2025 – June 2025

- Achieved sub-100ms latency for real-time multiplayer synchronization by architecting a full-stack solution using Node.js, Socket.IO, and WebRTC peer-to-peer streaming.
- Developed P2P Video and Audio Streaming capabilities, optimizing signaling pathways to enable low-latency, jitter-free communication between concurrent players.
- Deployed a scalable room-based matchmaking system on **Google App Engine (GCP)**, utilizing MongoDB for persistent state management and Express for secure RESTful API routing.

Language Sentiment Analyzer | *Python, PyTorch, Transformers*

May 2024

- Fine-tuned a BERT transformer model using PyTorch to analyze sentiment on 1M+ tweets, achieving 82% validation accuracy and optimizing data preprocessing pipelines with Pandas.
- Implemented data cleaning pipelines to tokenize and normalize large-scale text datasets, reducing noise to improve model convergence times.

Tab It! | *MongoDB, Express, React, Node.js*

Jan 2024 – March 2024

- Built a scalable RESTful API handling secure JWT authentication and complex receipt parsing logic, reducing data retrieval time by utilizing MongoDB indexing.
- Developed key user-facing components with React, including an interactive line-item assignment UI that allows multi-user bill splitting.

WORK EXPERIENCE

ANS Construction

Software Automation Intern

June 2023 – Sept 2023

Los Angeles, CA

- Reduced weekly financial reporting time by 40% by automating expense tracking and budget adherence analysis for construction projects using Python (Pandas) scripts to parse PDF invoices.

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, C/C++, Python, SQL, C#, OCaml, Java, Lisp, GLSL

Technologies: Three.js, WebGL, React, Node.js, Express, Socket.IO, MongoDB, GCP (App Engine), PyTorch, Git